

Project Description

1. System Overview & Data Volume

The Personal Finance Management System (PFMS) tracks personal finances through income/expense logging, investment portfolio analysis, budget monitoring, loan tracking, and AI-driven conversational guidance. The database consists of 16 core entities populated with 30,000+ total records, generated using Faker, a Python library for producing realistic synthetic data suitable for testing and prototyping.

Table	Minimum Rows
Transactions	20,000+
Investments (bonds, deposits, stocks) / Notification	4,000+
InvestmentTracking / Messages / UserNotification	1,500+ each
FinancialGoal / Budget / AIConversation / Users / Loan / Bank Account	1,000+ each

2. Queries

The document contains explanations for all the queries. The short names of the queries is written in the sql query document. Most of the functions are for general operations and analytics the user does, and other queries are for the admins to get the needed information for certain users.

Core Business Operations

Main operations are about:

User Registration

Balance Across All Bank Accounts

Recording Transactions

Portfolio Management

AI Assistant Messaging

Monthly Cash Flow (Spending vs. Earnings)

Budget Overrun Detection

Clear Explanations For Each of Queries is Written Below

Financial Goals

Query	Description
Goal completion percentage	Shows how close each goal is to its target, as a percentage

Query	Description
Overdue goals	Lists incomplete goals past their deadline, sorted by how many days overdue
Goals by priority with remaining amount	Shows incomplete goals ordered by priority and deadline, with how much is left to save

Investments

Query	Description
Investments worth more than ALL deposits	Finds investments whose amount exceeds every deposit made by a user
Tracking better than every bond	Finds any tracked investment outperforming all bond entries for a user
All active investments with type	Lists all currently active investments for a user, newest first
Users with active investments	Returns all users who have at least one active investment
Investments tracked in last 7 days	Shows investments that had a tracking entry recorded in the past week
Goals belong to verified accounts	Lists financial goals only for users with verified account status
Tracking negative entry	Finds investments that have at least one recorded loss in tracking history
Show all deposits	Displays deposit details including bank name, interest rate, and maturity date
Investment portfolio summary by type	Groups active investments by type, summing total invested and expected returns
Deposits maturing within 60 days	Lists deposits expiring soon, with days remaining until maturity
Best performing stock	Returns the single stock with the highest total profit for a user
All stocks with total portfolio value	Shows each stock's market value and unrealized profit/loss
Investments with negative change	Lists investment tracking entries where the value is declining
Tracking history for a specific investment	Shows the full value and percent change history for one investment
Average change % per investment type	Summarizes average, best, and worst performance per investment type

AI Conversations

Query	Description
Show all conversations	Lists all AI conversations for a user, newest first
Show pinned conversations	Returns only conversations the user has marked as pinned

Query	Description
Conversations with at least one message	Returns conversations that contain at least one message
Message count per conversation	Shows how many messages exist in each conversation
Last message in each conversation	Retrieves the most recent message from every conversation
Search conversations by keyword	Finds conversations where a keyword appears in any message
Show a conversation	Displays the full message thread for a specific conversation

Transactions & Bank Accounts

Query	Description
Most spent category	Finds the category where a user has spent the most money
Total balance across all bank accounts	Sums the balance of all accounts per user
Latest transactions (IN & OUT)	Returns the most recent completed transaction for each user
Last month's spendings	Totals all spending transactions within a given date range
Last month's earnings	Totals all earning transactions within a given date range
Transactions by category	Shows completed transactions belonging to a specific category
Cancelled transactions	Lists all transactions with a cancelled status
Show a specific bank account	Returns full details of one bank account by ID
Show all categories	Lists all transaction categories alongside any associated transactions
Show 30 latest notifications	Returns the 30 most recently sent notifications
Spendings exceeding average	Shows spending transactions above the user's own average spending amount

Budgets

Query	Description
Show all budgets	Lists all budgets for a user with category, limit, and dates
Show all active budgets	Same as above but filtered to currently active budgets only
Budgets over the limit	Identifies categories where total spending has exceeded the budget limit

Loans

Query	Description
Show all loans	Returns all loan records for a user
Show all loan payments	Lists loans joined with the user's bank account information
Current loans with due date and lender	Shows only active or overdue loans with key repayment details
Remaining loan amount	Estimates the remaining balance on each loan based on months elapsed and interest

Stocks & Bonds

Query	Description
Info of all bought stocks	Shows stock details including name, price, shares, and amount invested
Info of all owned bonds	Displays bond details including type, coupon frequency, and maturity date
Users with the highest balance	Ranks users by their total combined bank account balance
Users with multiple bank accounts	Lists users who hold more than one bank account, sorted by count

3. Triggers

TRIGGER 1: automatically creates a default checking account when a new user is added. It uses the newly inserted user's ID to insert a corresponding record into BankAccount. This ensures every user has an account without manual setup.

TRIGGER 2: updates the budget's consumed amount and adjusts account balances whenever a new transaction is inserted. It also checks budget limits and sends notifications if the spending exceeds the limit or reaches the alert threshold.

TRIGGER 3: checks the sender's bank account balance before a transaction is inserted. If the balance would go below zero after the transaction, it stops the operation and raises an error. This prevents users from making transactions without sufficient funds.

TRIGGER 4: updates the loan repayment status whenever a transaction linked to a loan is inserted. It calculates the total amount paid and remaining balance, and if the loan is fully paid, it marks it as "paid" and sends a notification. Otherwise, it sends an update showing the remaining loan balance.

TRIGGER 5: This trigger updates a financial goal's progress rate whenever the current amount changes. If the target is reached or exceeded, it automatically marks the goal as completed and sends a notification to the user.

FUNCTION 1: calculates a user's net cash flow over a given time period by summing all incoming and outgoing transactions. It then subtracts total expenses from total income to determine whether the user gained or lost money. If no transactions exist, it treats the values as zero to avoid null errors.

FUNCTION 2: calculates how much of a specific loan is still unpaid by subtracting all loan-related transactions from the original principal amount. If no payments exist, it treats the paid amount as zero to avoid null errors.

FUNCTION 3: computes a user's total balance across all their bank accounts by summing all account balances. If the user has no accounts or balances, it returns zero instead of null.

4. Indexes

The following indexes are created to improve query performance by allowing PostgreSQL to locate rows efficiently without performing full table scans.

1. idx_bankaccount_user on BankAccount(user_id)

This index speeds up queries that filter bank accounts by user_id. Without the index, PostgreSQL performs a sequential scan over all accounts. With the index, it can directly locate all accounts belonging to a specific user, significantly reducing lookup time.

2. idx_transactions_account on Transactions(account_id)

This index optimizes queries that retrieve transactions for a specific account_id. It avoids scanning the entire transactions table and instead enables fast access to only the relevant rows associated with a given account.

3. idx_transactions_date on Transactions(transaction_date)

This index improves performance for queries that filter transactions by date ranges (e.g., transactions after a certain date). PostgreSQL can efficiently locate a subset of rows using an index or bitmap index scan instead of scanning all records.

4. idx_investment_user on Investment(user_id)

This index accelerates queries that retrieve investment records for a specific user. It is particularly useful in user-focused applications where investment data is frequently accessed by user_id.

5. idx_conversation_user on AIConversation(interaction_id)

This index enables fast lookup of conversation records by interaction_id. If interaction_id is already defined as a primary key, PostgreSQL automatically creates an index, making this additional index redundant. Otherwise, it ensures efficient retrieval of specific interactions.